Which tag pair should an indie game use? First-Person × Multiplayer

MADT 7202 Mid-Term · Panjapol Ampornratana 6810424009 · Repo: github.com/hhej/steam-tag-whitespace

Steam is the largest PC game store. Players vote tags onto each game; the top tags drive store search.

1 The decision

An indie studio's creative director picks the next game's two lead tags; if no pair passed the selection rule, they would pick a crowded one (table, last row).

2 Why a network

Counts give crowding and reviews. Which uncrowded pairs join separate neighbourhoods, and through which tag, needs neighbours and paths, which counts lack.

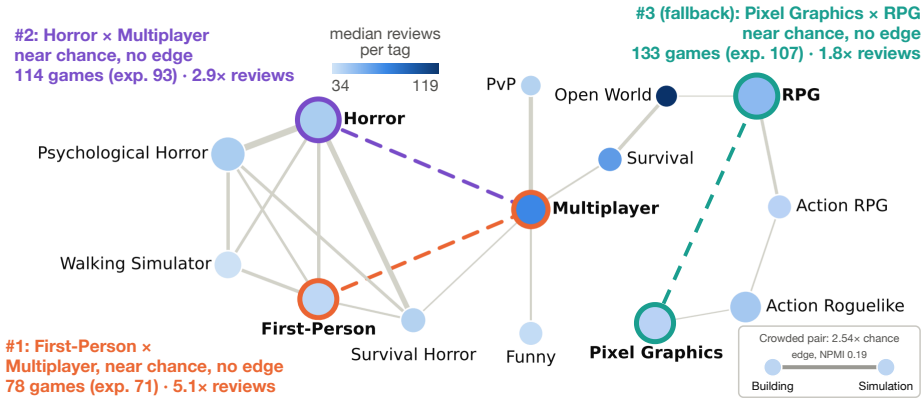
3 Data

Source	S1 FronkonGames steam-games-dataset (Hugging Face); S2 live Steam store.
Obtained	S1 download (Sep 2025). S2: 1,000 random S1 games (random seed 7202), scraped 2026-10-14.
Legal	S1 MIT licence. S2: robots.txt allows /app/, /appreviews/; 1 req/1.5 s; no personal data.
Volume	S1: 9,094 Indie games, 2023–24, ≥10 reviews; hold-out: 2,076 from Jan–Jun 2025. S2: 996 live pages.
Check	S1 vs S2: top-5 tag Jaccard 0.87; reviews Spearman ρ 0.98.
Bias	Favours games that sold; reviews = buyers, not unmet demand; late-2024 games had less time.

4 Graph construction

Node	A Steam tag. Undirected, weighted tag–tag projection of the game–tag bipartite graph.
Edge	Co-occurrence in a game's top 5 tags; suggests relatedness, not a stated relation.
Resolution	Co-op, Online/Local Co-Op, Local Multiplayer → Multiplayer (84% of Horror × Multiplayer games were co-op too).
Weight	NPMI: co-occurrence vs chance (0 = chance). By raw count, 9 of the top-10 edges touch the 5 commonest tags; by NPMI, 0.
Window	Top 5 tags per game, as the store page shows; with all 20, #1's reviews drop from 5.1× to 3.0×.
Excluded	"Indie"; Early Access, Free to Play, adult tags; on >25% of games: Singleplayer, Adventure, Action, Casual, 2D, 3D, Atmospheric, Exploration.
Pruning	Tag on ≥1.5% of games (136); edge needs NPMI ≥ 0.10 and ≥0.5% of games (45) with both tags. 439 tags → **80 nodes, 177 edges**.

5 The picture



Survival Horror links #1's tags: 2 steps apart, 5 without it. 14 of 80 tags: ranked pairs (bold), #1 and #2's top-2 neighbours, linking paths. Size = games, width = NPMI.

Selection rule: of pairs on ≥73 games (0.8%), keep those **not crowded** (games ≤ 1.5× chance; chance = games with A × games with B ÷ 9,094) with ≤20% **shared neighbours**; rank by median reviews ÷ 40.

Rank: pair	Games (chance)	Crowding	Shared neighbours	Median reviews
#1: **First-Person × Multiplayer**	78 (71)	1.10×	8%	204 (5.1×)
#2: **Horror × Multiplayer**	114 (93)	1.23×	8%	117 (2.9×)
#3: **Pixel Graphics × RPG**	133 (107)	1.25×	0%	70 (1.8×)
Crowded (box in figure): **Building × Simulation**	74 (29)	2.54×	8%	136 (3.4×)

6 Findings

- **#1** is near chance (NPMI 0.02, no edge) yet gets 5.1× the median reviews ($p < 0.001$, permutation test). Not horror: all horror games get 1.0×, #1's non-horror games 11.5×.
- **Survival Horror is the only link** between #1's two neighbourhoods (figure): the one tag to watch.
- **Mostly holds:** #1 at all 8 edge cutoffs and in 59/100 bootstraps; 3.35× without its top 5 games; on early-2025 games the analysis never saw, still uncrowded (1.14×) at 2.4× reviews.

7 Recommended action

- Week 0: pitch a first-person multiplayer game.
- Weeks 1–6: prototype. Week 7: Coming Soon page tagged First-Person and Multiplayer/Online Co-Op.
- Every quarter: re-run; watch #1 and Survival Horror crowding (section 8).

8 What would change my mind

First-Person is spreading (27% of 2025 games). If #1 passes 1.5× crowding, I switch to #2 while it stays under 1.5× (1.45× in 2025); if both are crowded, to #3.